

Mechatronics Folio and Journal

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1. Defining

1.1 Who is your client?

1.2 Describe the system you are making

An closed loop mechatronics autonomous vehicle built around a Raspberry Pi that runs modular Python software (drivers, perception with OpenCV/ML, planner, and actuator interface) and communicates with a NeZha I2C motor/servo controller to drive DC motors and servos. It uses a camera and sensors for vision-guided object detection and obstacle avoidance.

1.3 What are the specifications?

1.3.1 Function requirements

1.3.2 User input requirements

1.3.3 Output requirements

2. Planning

2.1 Materials required

Raspberry pi 5 8gb
Tracking sensor

2.2 Circuit diagram

2.3 UML Diagram

2.4 Structure Chart

2.5 Algorithms

3. Build Journal

Date and name	Goals	Image	Notes
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4. Unit Tests